HCML Monitoring Portal: database ERD

Hosts & Inventory

hosts

bigint	hostid	PK	
varchar(128)	host		technical name, map labels only
varchar(128)	name		visible name
int	status		0 monitored, 1 not monitored, 3 template
int	flags		0 plain, 2 prototype, 4 discovered
int	maintenance_status		1 in maintenance
text	description		

hstgrp

bigint	groupid	PK	
varchar(255)	name		
int	type		0 host group, 1 template group

hostid, cascade

host_inventory

bigint	hostid	PK,FK	no row when inventory is disabled
int	inventory_mode		0 manual, 1 automatic
varchar(128)	site_city		
text	location		
varchar(128)	poc_1_name		
varchar(128)	poc_1_email		
text	notes		

hostid, cascade

interface

bigint	interfaceid	PK	
bigint	hostid	FK	
int	main		1 the default interface
int	type		1 agent, 2 SNMP, 3 IPMI, 4 JMX
int	useip		1 connect by IP
varchar(64)	ip		
varchar(255)	dns		
int	available		0 unknown, 1 up, 2 down

hostid

hostid, cascade

host_tag

bigint	hosttagid	PK	
bigint	hostid	FK	
varchar(255)	tag		site, owner, criticality
varchar(255)	value		

groupid, cascade

hosts_groups

bigint	hostgroupid	PK
bigint	hostid	FK
bigint	groupid	FK

id, cascade

hostid

interfaceid

elementid when elementtype=0, no FK

API token & permissions

role

bigint	roleid	PK	
varchar(255)	name	UK	
int	type		1 user, 2 admin, 3 super admin

roleid, cascade

users

bigint	userid	PK
varchar(100)	username	UK
bigint	roleid	FK

usrgrp

bigint	usrgrpid	PK
varchar(64)	name	UK

creator_userid

userid, cascade

userid, cascade

usrgrpid, cascade

groupid, cascade

token

bigint	tokenid	PK	
varchar(64)	name		
bigint	userid	FK	every call runs as this user
varchar(128)	token	UK	SHA-512 of the secret
int	status		0 enabled, 1 disabled
int	expires_at		
bigint	creator_userid	FK	

users_groups

bigint	id	PK
bigint	usrgrpid	FK
bigint	userid	FK

rights

bigint	rightid	PK	
bigint	groupid	FK	the user group
bigint	id	FK	the host group
int	permission		0 deny, 2 read, 3 read-write

Items

items

bigint	itemid	PK	
bigint	hostid	FK	
bigint	interfaceid	FK	
varchar(255)	name		
varchar(2048)	key_		
int	value_type		picks the history and trend table
varchar(255)	units		
int	status		0 enabled, 1 disabled

itemid, cascade

item_rtdata

bigint	itemid	PK,FK	
int	state		0 normal, 1 not supported
varchar(2048)	error		

itemid

item_tag

bigint	itemtagid	PK	
bigint	itemid	FK	
varchar(255)	tag		link_group, link_role
varchar(255)	value		

Maps

sysmaps

bigint	sysmapid	PK	
varchar(128)	name	UK	
int	width		
int	height		
bigint	userid	FK	the owner

sysmapid, cascade

sysmaps_elements

bigint	selementid	PK	
bigint	sysmapid	FK	
bigint	elementid		no FK, elementtype says what it points at
int	elementtype		0 host, 1 map, 2 trigger, 3 group, 4 image
varchar(2048)	label		macros stored unresolved
int	x		
int	y		

sysmapid, cascade

selementid1, cascade

selementid2, cascade

sysmaps_links

bigint	linkid	PK	
bigint	sysmapid	FK	
bigint	selementid1	FK	
bigint	selementid2	FK	
varchar(6)	color		hex without the #

userid

itemid

userid, cascade

Triggers, events & problems

triggers

bigint	triggerid	PK	
varchar(2048)	expression		refers to items through functions
int	manual_close		1 Close permitted
varchar(255)	opdata		template behind problem opdata

triggerid

objectid when object=0, no FK

functions

bigint	functionid	PK
bigint	itemid	FK
bigint	triggerid	FK

events

bigint	eventid	PK	
int	source		0 triggers, 4 services
int	object		0 trigger, 4 item, 5 LLD rule, 6 service
bigint	objectid		no FK, object says what it points at
int	clock		
int	value		1 problem, 0 recovery
int	acknowledged		set by event.acknowledge
varchar(2048)	name		
int	severity		0 to 5

objectid when object=0, no FK

c_eventid, cascade

r_eventid, cascade

eventid, cascade

eventid, cascade

r_eventid, cascade

cause_eventid

event_recovery

bigint	eventid	PK,FK	the problem event
bigint	r_eventid	FK	the event that resolved it
bigint	c_eventid	FK	the correlation event, if any

problem

bigint	eventid	PK,FK	
int	object		
bigint	objectid		no FK
int	clock		
bigint	r_eventid	FK	NULL while open
int	r_clock		
bigint	cause_eventid	FK	set when this is a symptom
varchar(2048)	name		
int	acknowledged		set by event.acknowledge
int	severity		

eventid, cascade

problem_tag

bigint	problemtagid	PK
bigint	eventid	FK
varchar(255)	tag	
varchar(255)	value	

History & trends

itemid, value_type 2, no FK

history_log

bigint	itemid	PK	no FK to items
int	clock	PK	
int	ns	PK	
text	value		

itemid, value_type 0, no FK

history

bigint	itemid	PK	no FK to items
int	clock	PK	
int	ns	PK	
double	value		

itemid, value_type 3, no FK

history_uint

bigint	itemid	PK	no FK to items
int	clock	PK	
int	ns	PK	
bigint	value		

itemid, value_type 1, no FK

history_str

bigint	itemid	PK	no FK to items
int	clock	PK	
int	ns	PK	
varchar(255)	value		

itemid, value_type 0, no FK

trends

bigint	itemid	PK	no FK to items
int	clock	PK	start of the hour
int	num		values in that hour
double	value_min		
double	value_avg		
double	value_max		

itemid, value_type 4, no FK

history_text

bigint	itemid	PK	no FK to items
int	clock	PK	
int	ns	PK	
text	value		

itemid, value_type 3, no FK

trends_uint

bigint	itemid	PK	no FK to items
int	clock	PK	start of the hour
int	num		values in that hour
bigint	value_min		
bigint	value_avg		
bigint	value_max		

eventid, cascade

eventid, cascade

matched on tag and value by the server

Acknowledge writes

acknowledges

bigint	acknowledgeid	PK	
bigint	userid	FK	the API token user
bigint	eventid	FK	
int	clock		
varchar(2048)	message		
int	action		bitmask 1 close, 2 ack, 4 message

task

bigint	taskid	PK	
int	type		1 close problem, 4 acknowledge
int	status		
int	clock		

acknowledgeid, no FK

taskid, cascade

acknowledgeid, no FK

taskid, cascade

task_close_problem

bigint	taskid	PK,FK	
bigint	acknowledgeid		no FK

task_acknowledge

bigint	taskid	PK,FK	
bigint	acknowledgeid		no FK

Services

services

bigint	serviceid	PK	
varchar(128)	name		
int	status		-1 OK, else a severity
int	algorithm		
int	sortorder		
text	description		

serviceid, cascade

service_problem

bigint	service_problemid	PK	
bigint	eventid	FK	a problem row, not an event
bigint	serviceid	FK	
int	severity		

servicedownid, cascade

serviceupid, cascade

services_links

bigint	linkid	PK	
bigint	serviceupid	FK	the parent
bigint	servicedownid	FK	the child

serviceid, cascade

service_tag

bigint	servicetagid	PK	
bigint	serviceid	FK	
varchar(255)	tag		
varchar(255)	value		

serviceid, cascade

service_alarms

bigint	servicealarmid	PK	
bigint	serviceid	FK	
int	clock		
int	value		status from this clock on

serviceid, cascade

service_problem_tag

bigint	service_problem_tagid	PK	
bigint	serviceid	FK	
varchar(255)	tag		
int	operator		
varchar(255)	value		

matched on tag and value, no key

SLA

sla

bigint	slaid	PK	
varchar(255)	name	UK	
int	period		0 daily to 4 annually
double	slo		target percent
int	effective_date		
varchar(50)	timezone		
int	status		0 disabled, 1 enabled
text	description		

slaid, cascade

slaid, cascade

slaid, cascade

sla_excluded_downtime

bigint	sla_excluded_downtimeid	PK
bigint	slaid	FK
varchar(255)	name	
int	period_from	
int	period_to	

sla_schedule

bigint	sla_scheduleid	PK
bigint	slaid	FK
int	period_from	
int	period_to	

sla_service_tag

bigint	sla_service_tagid	PK
bigint	slaid	FK
varchar(255)	tag	
int	operator	
varchar(255)	value	